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Course Project Specification

BIRD

Birzeit Integrated Router Design

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BIRD – Birzeit Integrated Router Design

Functional Specification

Contents

1 Overview	2
2 Packet Data and Control Signals	2
2.1 Packet Data Stream	2
2.2 Configuration Signal (cfg)	2
2.3 Sampling Rule	2
3 Interface Protocol	2
3.1 Transfer Rule	2
3.2 Stability Rule	3
4 Signal Interfaces	3
4.1 Global Signals	3
4.2 Status Output Interface	3
4.3 Input Interface (Producer → BIRD)	3
4.4 Local Output Interface	3
4.5 Remote Output Interface	3
5 Configuration Word (cfg) Format	3
6 Local Traffic Processing	4
7 Remote Traffic Processing	4
7.1 Fragment Rules	4
7.2 Accumulation and Reassembly	4
7.3 Remote Output Format	5
8 Silent Packet Drop Policy	5
8.1 Drop Conditions	5
8.2 Drop Counter Behavior	5
9 Reset Behavior	5
10 Notes for Verification	6

1 Overview

BIRD (Birzeit Integrated Router Design) is a packet-based routing block that receives traffic from a single input interface and routes it to one of two output interfaces: **local** or **remote**. Routing decisions and packet processing behavior are determined by a 32-bit configuration signal (**cfg**) that accompanies packet data on the input interface.

BIRD supports two traffic types:

- **Local traffic**: forwarded directly to the local output interface.
- **Remote traffic**: received as multiple fragments, accumulated, reordered, merged, and transmitted as a single packet.

BIRD does not signal errors. Any invalid or malformed packet is silently dropped and recorded internally.

2 Packet Data and Control Signals

2.1 Packet Data Stream

The packet data stream transferred on **data_in** consists of:

Payload (1--255 bytes)

CRC16 (16 bits)

- Payload data is transferred at 8 bits per cycle.
- CRC16 covers only the payload bytes.
- CRC16 is transferred immediately after the payload.

2.2 Configuration Signal (**cfg**)

The **cfg** signal is a 32-bit **sideband control signal** and is **not part of the packet data stream**.

- **cfg** describes the fragment currently being transferred.
- **cfg** must be valid and stable during the transfer of each fragment.
- For remote traffic, **cfg** is provided with **each fragment**.
- For local traffic, **cfg** applies to the entire packet.

2.3 Sampling Rule

BIRD samples the **cfg** signal on the same clock cycle as the first payload byte of each fragment, when **in_vld** and **in_rdy** are asserted.

3 Interface Protocol

All interfaces use a valid/ready handshake.

3.1 Transfer Rule

A transfer occurs on a rising edge of **clk** when:

$$\text{vld} = 1 \wedge \text{rdy} = 1$$

3.2 Stability Rule

When `vld = 1` and `rdy = 0`, all associated data and control signals must remain stable.

4 Signal Interfaces

4.1 Global Signals

Signal	Width	Direction	Description
<code>clk</code>	1	Input	System clock (rising-edge triggered)
<code>rst_n</code>	1	Input	Active-low reset

4.2 Status Output Interface

Signal	Width	Direction	Description
<code>drop_cnt</code>	16	Output	Number of packets dropped (wrap-around counter)

4.3 Input Interface (Producer $\rightarrow$ BIRD)

Signal	Width	Direction	Description
<code>in_vld</code>	1	Input	Input data valid
<code>in_rdy</code>	1	Output	BIRD ready to accept input
<code>data_in</code>	8	Input	Payload and CRC data stream
<code>cfg</code>	32	Input	Sideband configuration for current fragment

4.4 Local Output Interface

Signal	Width	Direction	Description
<code>local_vld</code>	1	Output	Local output valid
<code>local_rdy</code>	1	Input	Local consumer ready
<code>data_local</code>	8	Output	Local payload data

4.5 Remote Output Interface

Signal	Width	Direction	Description
<code>remote_vld</code>	1	Output	Remote output valid
<code>remote_rdy</code>	1	Input	Remote consumer ready
<code>data_remote</code>	32	Output	Remote packet data stream

5 Configuration Word (`cfg`) Format

The `cfg` signal is a 32-bit sideband control word with the following format:

Bits	Field	Width	Description
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[0]	TRAFFIC_TYPE	1	0 = Local traffic, 1 = Remote traffic
[7:1]	Reserved	7	Must be 0
[15:8]	PAYLOAD_LEN	8	Payload length in bytes (1–255)
[20:16]	FRAG_NUM	5	Fragment number within the packet (1–31)
[23:21]	Reserved	3	Must be 0
[28:24]	SEQ_NUM	5	Packet sequence number (1–31). 0 is invalid
[31:29]	Reserved	3	Must be 0

6 Local Traffic Processing

A fragment is classified as local traffic when:

$$\text{cfg}[0] = 0$$

- Local traffic consists of a single fragment only.
- Payload bytes are forwarded directly to the local output interface.
- No buffering, accumulation, or reordering is performed.
- FRAG_NUM shall be equal to 1.
- SEQ_NUM identifies the packet but has no functional impact on local routing.
- The input CRC16 is forwarded unchanged.

7 Remote Traffic Processing

A fragment is classified as remote traffic when:

$$\text{cfg}[0] = 1$$

7.1 Fragment Rules

- A remote packet consists of one or more fragments.
- Each fragment carries a FRAG_NUM field indicating its position within the packet.
- All fragments belonging to the same packet share the same SEQ_NUM.
- Valid fragment numbers are $1, 2, \dots, N$, where N is determined by received fragments.
- Fragments may arrive out of order.
- Only one remote packet (identified by SEQ_NUM) may be accumulated at a time.

7.2 Accumulation and Reassembly

1. Each fragment payload is buffered and indexed by FRAG_NUM.
2. All fragments sharing the same SEQ_NUM are considered part of the same packet.
3. When all required fragments are received, fragments are reordered based on FRAG_NUM.
4. Payloads are concatenated to form a merged payload.
5. A new CRC16 is generated over the merged payload.

7.3 Remote Output Format

The remote output interface transmits:

Merged Payload
CRC16

The `cfg` signal is not transmitted on the remote output interface.

8 Silent Packet Drop Policy

BIRD does not signal errors. Invalid packets are silently dropped and counted.

8.1 Drop Conditions

A packet is dropped if any of the following occur:

- `SEQ_NUM == 0`
- `FRAG_NUM == 0`
- `PAYLOAD_LEN` is outside the range 1–255
- Any reserved bits in `cfg` are non-zero
- A fragment arrives with a mismatched `SEQ_NUM` while another packet is being accumulated
- A required fragment for a packet is missing
- A fragment with `FRAG_NUM == 1` arrives while a previous packet with a different `SEQ_NUM` is still incomplete

8.2 Drop Counter Behavior

- `drop_cnt` increments by one for each packet that is dropped.
- The counter increments once per packet, regardless of the number of fragments received.
- `drop_cnt` is a modulo- 2^{16} counter and wraps around on overflow.
- `drop_cnt` is not affected by successful packet transmission.

When a drop condition is detected, all buffered data for the affected packet is immediately discarded.

9 Reset Behavior

When `rst_n = 0`:

- All valid outputs are deasserted.
- All internal buffers and fragment state are cleared.
- Any packet in progress is discarded.
- `drop_cnt` is cleared to zero.

Normal operation resumes after reset deassertion.

10 Notes for Verification

- Verify correct classification of local versus remote traffic.
- Verify handshake stability under backpressure.
- Verify correct use of `FRAG_NUM` and `SEQ_NUM`.
- Verify fragment reordering and merged payload correctness.
- Verify silent drop behavior under all listed conditions.
- Verify correct incrementation and wrap-around behavior of `drop_cnt`.
- Verify CRC16 regeneration for remote packets.